



Quantum Key Distribution

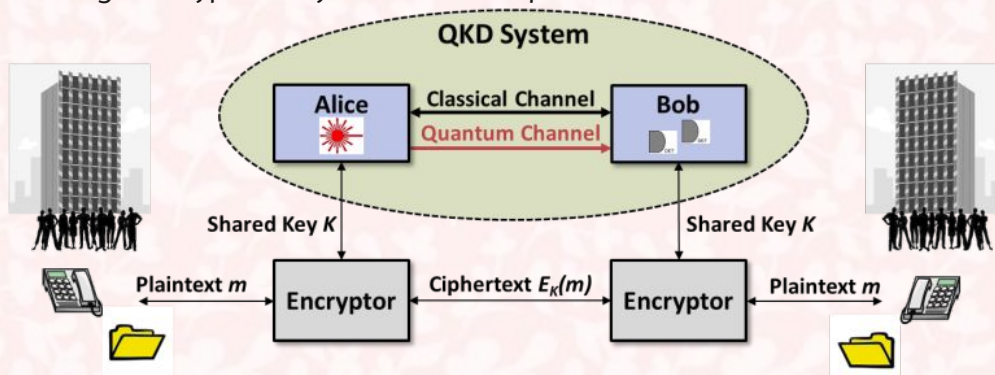
Aarya Raina





What is Quantum Key Distribution?

Quantum Key Distribution (QKD) is a method of secure communication that uses principles of quantum mechanics to exchange encryption keys between two parties.



FUNDAMENTAL IDEA: To leverage the properties of quantum mechanics, such as the uncertainty theorem and the no cloning theorem, to facilitate a form of communication that is secure against eavesdropping parties.



What is a one-time pad?

A one-time pad refers to a cryptographic key that is used only once for encrypting and decrypting a single message.

- In quantum, they use Qubits, which is able exist in the superposition of states. Through this, they use the XOR (CNOT) Gate. It generally flips the target qubit (the second one) if and only if the control qubit (the first one) is in a state of $|1\rangle$

ENCRYPT

$$\begin{array}{rcl} & \oplus & \\ \text{00110101} & \text{Plaintext} & \\ \text{11100011} & \text{Secret Key} & \\ \hline = & \text{11010110} & \text{Ciphertext} \end{array}$$

DECRYPT

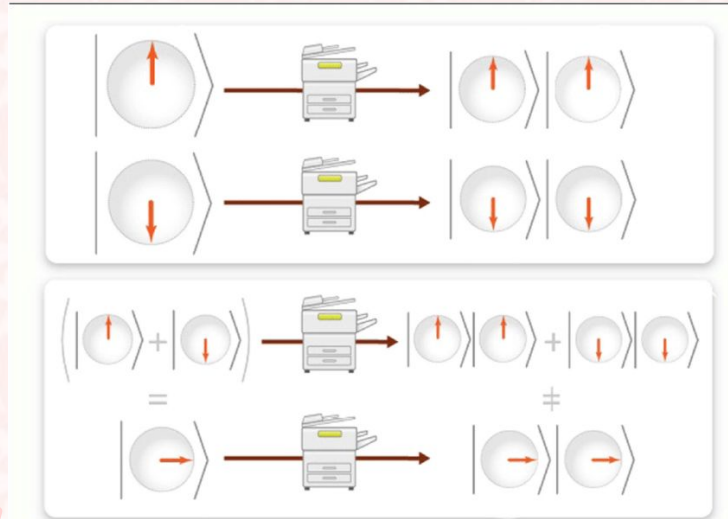
$$\begin{array}{rcl} & \oplus & \\ \text{11010110} & \text{Ciphertext} & \\ \text{11100011} & \text{Secret Key} & \\ \hline = & \text{00110101} & \text{Plaintext} \end{array}$$





What is the no cloning theorem?

- This theorem asserts two things, that there is no perfect copy, meaning that it is impossible to create an exact copy of an arbitrary unknown quantum state, and its implications, meaning that there is no universal quantum operation that can clone a quantum state without altering or disturbing its original quantum state.
- In theory, this means that nobody can copy a qubit since it will disturb the original quantum state due to the no cloning theorem.





How does it work?

This is Alice. Alice wants to send an encrypted message to Bob.



They agree to use Quantum Key Distribution to encrypt the data. In this example, They choose the vertical polarized photon to represent 1, and a horizontally polarized photon to represent 0

Alice will send encoded photons through a quantum channel to Bob. The channel is normally optic fiber for ground based communications, or free space for short distance free space communication.

This is Bob.





How does it work?

Alice begins to transmit
either horizontal or
vertical polarized photons
to Bob on a random basis



Alice and Bob compare bases through
a classical communication channel.
Through this they compare and
measure for the degree of error.
They go through this procedure until
they end at a suitable degree of
error.

Upon receiving the photon, Bob
chooses one of the two bases
randomly to measure the photons
polarization. He records each
outcome





How does it work?

However, when he and Alice compare through classical means, they find they have a high error! This means that there was an eavesdropper, and they should redo the procedure until they have a low error.



BUT! There seems to be an eavesdropper, we'll call her Eve. she intercepted the photon and sent a differently polarized one to Bob.



Once they have a secure one-time pad between the two of them, they're able to send messages in an encrypted server without worrying about eavesdroppers!





What if they wanted to try another way?

QKD would have to be implemented with entanglement, which normally involved photon pairs to establish a secure cryptographic key.

How would they do that?

- To start- Alice would need a specialized photon source that generates pairs of photons that are entangled. (Entanglement is when two particles, in this case photons, link together in a certain way no matter how far apart they are in space. Their state remains the same.)
- The entangled photon pair is normally prepared in a quantum state such as the maximally entangled Bell state:

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Where $|0\rangle$ and $|1\rangle$ represent the quantum states of polarization and $|00\rangle$ and $|11\rangle$ denote the two possible pairs of outcomes for photons in an entangled state.

- Similar to the BB84 protocol (as just shown), they both use a measurement of 1 or 0 based on the polarization. However, in this example the photons they have are the opposite of each other, such as $|H\rangle$ and $|V\rangle$ or $|A\rangle$ and $|D\rangle$





Quick Overview, what does the Bell state mean?

A Bell state, named after physicist John S. Bell, is a specific type of entangled quantum state involving 2 qubits. Entanglement means that one qubit is correlated with another and the state of one is dependent on the state of the other. (Think of Schrodinger's cat)

$$\text{Equation: } |\Phi+\rangle = 1/\sqrt{2} (|00\rangle + |11\rangle)$$

Other Bell states can be derived from $|\Phi+\rangle$ by applying different quantum gates (ex. Z, H, X)

$$|\Phi-\rangle = 1/\sqrt{2} (|00\rangle - |11\rangle)$$

$$|\Psi+\rangle = 1/\sqrt{2} (|01\rangle + |10\rangle)$$

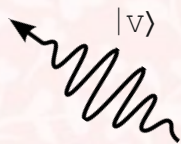
$$|\Psi-\rangle = 1/\sqrt{2} (|01\rangle - |10\rangle)$$



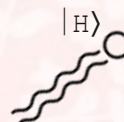


How does it work?

- Both Bob and Alice measure the polarization in the same way Bob did for BB84
 - Due to entanglement, both Bob's and Alice's photons are correlated. So, if Alice gets a 1, Bob should get a zero.
 - Both communicate the bases and repeat the procedure until a low margin of error, or if Eve was spotted. (She was the eavesdropper)



This is the photon transmitter





Pros and Cons

Pros

- Unconditional Security
- Future-proof against quantum computers (resistant against attacks from quantum computers)
- Applications in secure communication

Cons

- Technical challenges
- Distance limitations
- Key rate and efficiency
- Practical deployments

